

Hockey Player Robomaster_EP

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Abstract—This project introduces the design of an autonomous RoboMaster EP robot that plays hockey using ROS2 and motion capture feedback. The robot had to pick up a hockey stick by itself and then use the robotic arm and gripper to pick up the hockey stick, move to a puck, place itself behind the hockey puck and push the puck to a designated goal. A finite state machine design was created to handle the order of execution of these tasks. Robot navigation was achieved using approximate linearization of the unicycle dynamics by introducing a virtual control point p ahead of the robot. This allowed to use a simple proportional position controller for way point tracking. To make sure that the robot remained safe to operate in an environment of multiple robots, the Control Barrier Functions (CBFs) were added as a safety layer, which transformed potentially unsafe control commands generated by the nominal navigation controller. The controller thus enabled the robot to keep a safe distance from other robots in near proximity while executing the assigned task of performing the hockey manoeuvre. The experimental testing showed successful acquisition of a stick, object manipulation, approach to the puck and collision avoidance in the workspace.

Index Terms—RoboMaster EP, Approximate Linearization, CLF-CBF-QP-Based Controller, ROS2, Mobile Robotics, Autonomous Navigation, Unicycle

I. PROPOSED APPROACH

A. (T0) System Architecture

The autonomous hockey-playing robot was implemented as a finite state machine (FSM) in ROS2. The FSM is programmed to switch between a series of pre-defined states to coordinate navigation, manipulation, collision avoidance and scoring behaviors. The whole system architecture is shown as:

GET_CHECKPOINT → GET_STICK → GRAB_STICK →
GET_GOAL_FAR → GET_GOAL_NEAR →
SWING_SHOT → DONE

The robot first moves to a check-point near hockey stick area, and then performs a precise final approach. Once the hockey stick is retrieved, the robot moves to the goal waypoints where it sets-up and align for shooting maneuver. Finally, a rotational swing is performed to hit the puck into the goal. A Control Barrier Function (CBF) safety filter is used to monitor the nearby robots and adjust the nominal navigation commands to avoid a collision when constraints are triggered during all stages of the navigation.

The finite state machine logic was implemented in the `timer_callback()` function, and individual state behaviors were implemented in separate functions.

B. (T1) Navigation Controller; Approximate Linearization

The RoboMaster EP is modeled as a unicycle robot whose kinematics are described by

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega$$

To simplify controller design, a virtual point p located a distance l ahead of the robot was used.

$$p_x = x + l \cos \theta, \quad p_y = y + l \sin \theta$$

The desired point tracking was achieved using a proportional position feedback controller

$$u_x = k_p(x_d - p_x), \quad u_y = k_p(y_d - p_y)$$

The point velocities were converted into linear and angular velocity commands through approximate linearization

$$v = u_x \cos \theta + u_y \sin \theta, \quad \omega = \frac{-u_x \sin \theta + u_y \cos \theta}{L_s}$$

where L_s represents the steering gain `self.STEER_GAIN_L` parameter used in the implementation. This controller was used during the `GET_CHECKPOINT`, `GET_STICK`, `GET_GOAL_FAR`, and `GET_GOAL_NEAR` states.

C. (T2) Picking Up a Stick

Once the robot was at the pick-up point, it uses the on-board arm and the gripper to pick up the hockey stick. In the absence of testing time with the RoboMaster EP, a custom arm simulator was created using the measured dimensions of the robot arm and gripper. The simulator was used during development to verify that the gripper could reach the hockey stick before testing it on physical robot. The end-effector position was computed using the forward kinematics equations

$$x_e = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2), \quad y_e = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

The stick pickup strategy consist of extending the arm toward the stick, closing the gripper, lifting the stick, reversing the robot to create clearance and arm down to rest position. A timer-based state machine was used to coordinate these actions.

The pickup procedure was used in the `grab_the_stick()` function, while arm positioning commands were generated through the `arm_controller()` function.

D. (T3) Navigating to a Known Puck Location

After the robot successfully retrieved the hockey stick, and then moved towards a Waypoints predefined from known goal location. Instead of going right to the puck, a multi-stage navigation strategy was used to set the robot in a good shooting configuration. The first waypoint, called the far goal waypoint, located at a safe distance from the goal and served as an intermediate target. Once the robot reached this location, it transitioned to a second waypoint, called the near goal waypoint. This waypoint positioned the robot closer to the puck while aligning the hockey stick with the desired shooting position. As suggested by the Professor.

The waypoint positions were computed using fixed offsets from the known goal location available through positioning system. This staged navigation approach improved positioning accuracy and reduced the likelihood of large steering corrections near the final shooting point. The waypoint navigation behaviors were implemented in the `go_to_goal_far()` and `go_to_goal_near()` functions.

E. (T4) Shooting a Puck into a Goal at a Known Location

Once the robot reached the near-goal waypoint, the robot was positioned in a shooting configuration relative to the puck and goal. Instead of continuously pushing the puck, a swing-shot strategy was adopted to transfer momentum to the puck and direct it toward the goal. The shooting maneuver was implemented as a dedicated state called `SWING_SHOT`. During this state, the robot temporarily halted its translational motion and executed a rapid rotational movement using a constant angular velocity command. The hockey stick, attached to the front of the robot, acted as a striking mechanism that transferred momentum to the puck.

This approach was selected because it simplified the final scoring action and reduced the sensitivity of the controller to localization errors near the goal. The waypoint-based positioning stages were implemented in the `go_to_goal_far()` and `go_to_goal_near()` functions, while the shooting action was implemented in the `swing_shot()` function.

F. (T5) Avoiding Obstacles Placed in the Environment

To ensure the safe operation in a multi-robot environment, a Control Lyapunov Function (CLF) and Control Barrier Function (CBF) based controller was integrated with the nominal navigation controller. The approximate linearization controller described in Task 1 generates a nominal control input (CLF)

$$u_{nom} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}$$

Which drives the robot toward its objective. Eventhough this controller achieves waypoint tracking, it does not explicitly consider obstacle avoidance and may generate unsafe trajectories when other robots are present nearby. To enforce safety, a Control Barrier Function was defined for each nearby robot as

$$h_i = (p_x - x_i)^2 + (p_y - y_i)^2 - d_{safe}^2$$

where (p_x, p_y) is the virtual point associated with the controlled robot, (x_i, y_i) is the position of the obstacle robot, and d_{safe} is the minimum allowable separation distance. The barrier function derivative is

$$\dot{h}_i = 2(p_x - x_i)u_x + 2(p_y - y_i)u_y$$

Safety is assured whenever the following CBF constraint is satisfied

$$\dot{h}_i + \gamma h_i \geq 0$$

where $\gamma > 0$ is a tuning parameter controlling the rate at which the robot moves away from unsafe regions. The resulting CLF-CBF-QP optimization problem

$$u^* = \arg \min_u \|u - u_{nom}\|^2$$

subject to

$$2(p_x - x_i)u_x + 2(p_y - y_i)u_y + \gamma h_i \geq 0$$

for all nearby robots. Instead of using an external QP solver, a projection-based safety filter was implemented. When an obstacle was detected, the controller modified the nominal control input to maintain a safe distance while continuing toward the target. A small tangential motion was added to help the robot move around obstacles instead of stopping in front of them. To reduce computation, only nearby robots were considered during collision avoidance. The collision avoidance framework was used in the `get_nearby_robots()`, `compute_barrier_derivatives()`, and `solve_clf_cbf_qp()` functions and remained active throughout all navigation states.

II. RESULTS

A. (T1) Navigating to a Known Stick Pick-Up Location

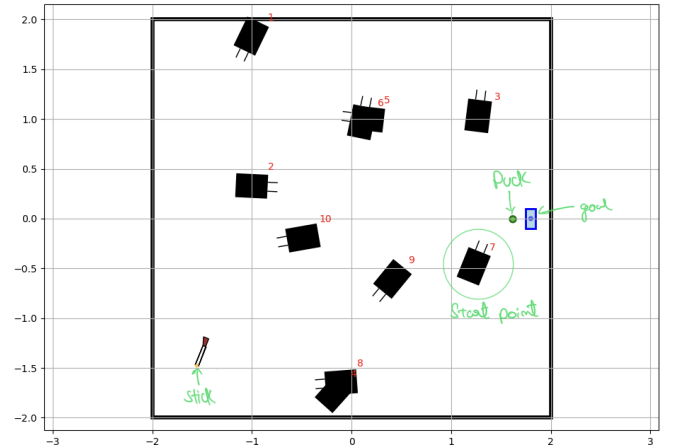


Fig. 1. Initial environment configuration

The first objective of the project is to navigate the RoboMaster EP from its initial position to a predefined hockey stick pick-up location. Figure 1 shows the initial configuration,

including the robot, hockey stick, goal, puck, and neighboring obstacle robots. To simplify development and testing, the provided simulation environment was modified to include visualization of the hockey objects. These additions allowed rapid validation of the navigation and collision-avoidance algorithms before testing them on the physical RoboMaster.

The robot successfully reached the desired pick-up location using the checkpoint-based navigation strategy described in Section I. By first moving toward an intermediate waypoint and then performing a final approach maneuver, the robot was able to align itself with the hockey stick having some Offset while avoiding unnecessary steering corrections (avoiding oscillatory behavior).

During the experimental phase, the navigation controller was operating at a control frequency of 50Hz with a proportional gain of $k_p = 1.2$. To ensure smooth and stable operation and prevent excessive actuator commands, the linear velocity and angular velocity was limited to

$$-0.5 \leq v \leq 0.5 \text{ m/s} \quad \& \quad -1.0 \leq \omega \leq 1.0 \text{ rad/s}$$

which provided smooth motion and reduced oscillatory behavior during waypoint tracking. Throughout the navigation process, the approximate linearization controller generated smooth motion commands. Experimental observations showed that the robot was able to consistently reach the desired pick-up location, providing a reliable starting configuration for the stick acquisition task described in Task 2.

B. (T2) Picking Up a Stick

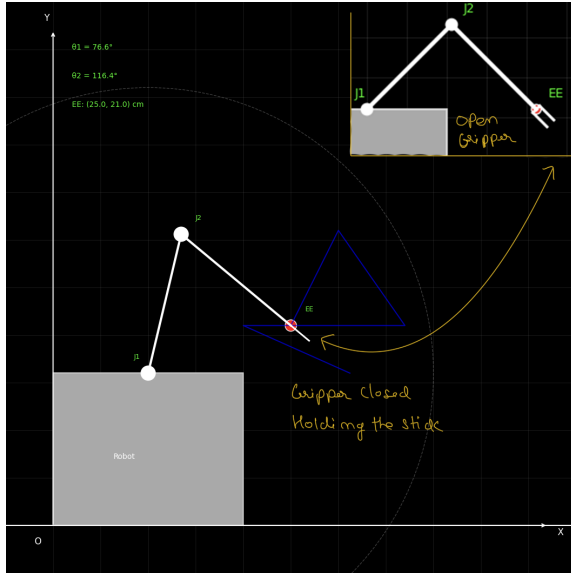


Fig. 2. Manipulator trajectory

The second objective was to successfully acquire the hockey stick using the RoboMaster EP's arm and gripper. Since testing time with the physical robot was limited, a custom simulator was developed using the measured dimensions of the RoboMaster EP to visualize the manipulator motion and

verify gripper placement before hardware testing. Figure 2 Smaller arm image on the top right, which shows the initial manipulator configuration with the gripper in the open state. This visualization was used to verify that the target hockey stick location was accessible with the arm. Figure 2 shows the bigger arm illustrating the arm trajectory used during the pickup operation. Experimental testing demonstrated successful execution of the complete pickup sequence. The stick pickup sequence consisted of four steps:

- 1) Extend the robotic arm toward the hockey stick.
- 2) Close the gripper to grasp the stick.
- 3) Lift the stick by raising the arm.
- 4) Reverse the robot to make clearance for next task.

C. (T3) Navigating to a Known Puck Location

After acquiring the hockey stick, the robot navigated to the shooting region using two intermediate waypoints generated from the known goal location. A far-goal waypoint and a near-goal waypoint were used to gradually position the robot in a favorable shooting configuration. The approximate linearization controller guided the robot between these waypoints while maintaining smooth motion and stable tracking performance. Real-time position feedback obtained from the positioning system continuously updated the control inputs and reduced navigation error.

The waypoint-based strategy positioned the robot for the final shooting maneuver described in Task 4.

D. (T4) Shooting a Puck into a Goal at a Known Location

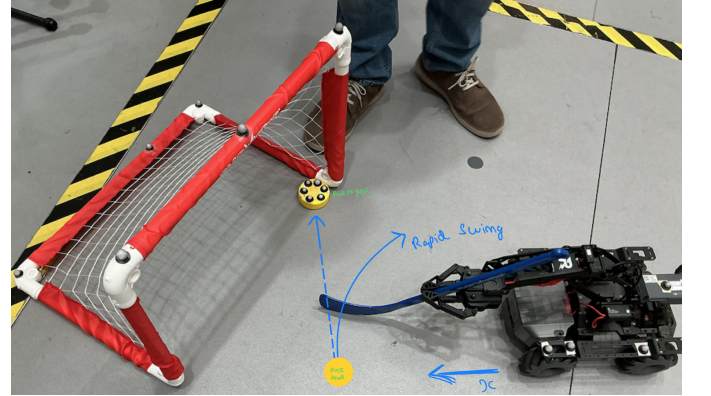


Fig. 3. RoboMaster EP setup used for the swing-shot experiment.

After reaching the near-goal waypoint, the robot executed a swing-shot maneuver to direct the puck toward the goal. Instead of continuously pushing the puck, the robot positioned itself beside the puck and applied a rapid rotational motion using the attached hockey stick. Figure 3 illustrates the shooting concept used in the final implementation. The robot first positioned the stick near the puck and then executed a high-speed rotational swing, transferring momentum from the stick to the puck. This resulted in the puck moving from its initial position toward the goal.

The swing-shot maneuver was successfully implemented and validated on the physical RoboMaster EP platform. And because the maneuver was developed and tested directly on the robot, detailed trajectory data was not recorded during the final demonstrations.

E. (T5) Avoiding Obstacles Placed in the Environment

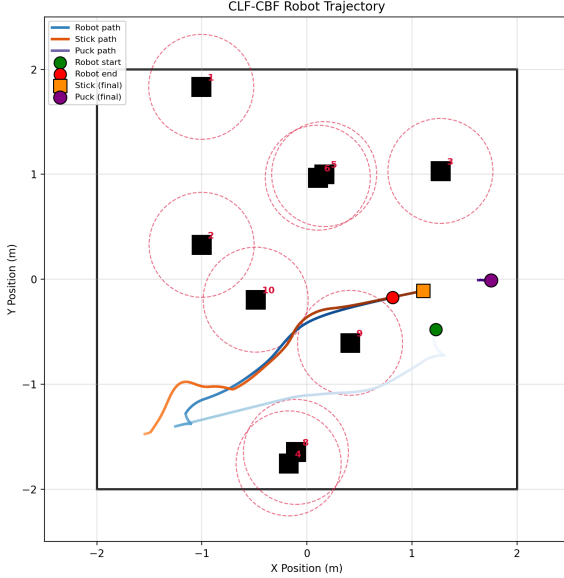


Fig. 4. Experimental trajectory generated by the CLF-CBF-QP

The CLF-CBF-QP-based controller was tested in the presence of multiple neighboring robots acting as obstacles. The CLF controller generated control commands that guided the robot toward its navigation target, while the CBF controller monitored the distance to nearby robots and enforced the safety constraints. The nominal CLF controller occasionally gave negative CBF values, indicating that the robot was moving toward an unsafe region. In these situations, the QP safety filter modified the control input and generated a safe command.

Figure 4 shows the resulting robot trajectory. The black squares represent neighboring robots, while the dashed red circles indicate the safety regions around each obstacle. The robot was able to navigate around obstacles and continue toward its objective without colliding with the obstacles. These results demonstrate that the CLF-CBF-QP-based controllers combined waypoint tracking and collision avoidance during navigation.

III. DISCUSSION

The proposed controller successfully integrated navigation, manipulation, and obstacle avoidance into a single autonomous hockey playing framework. Approximate linearization provided a simple and effective method for controlling the RoboMaster EP, while the finite-state machine architecture simplified the coordination of the different project tasks. The

custom arm simulator proved valuable during development by allowing manipulator motions to be verified before deployment on the physical robot. This reduced debugging time and improved confidence in the stick pickup procedure despite limited access to the hardware, such as the camera and encoders. The CLF-CBF-QP-based controllers successfully combined waypoint tracking and collision avoidance. Experimental results demonstrated that the robot could continue moving toward its objective while maintaining a safe distance from neighboring robots.

One limitation of the proposed approach is that the swing-shot strategy was tuned experimentally and therefore remains sensitive to the robot position, orientation, and puck placement. In addition, detailed trajectory data from the final demonstrations was not recorded, as the primary focus during the final testing sessions was to achieve a fully functional end-to-end system within the limited time available. Furthermore, the RoboMaster EP was modeled as a unicycle robot, which simplified the controller design but did not fully utilize the omnidirectional capabilities provided by its omniwheel. Lateral motion could be used in the future to enhance task performance, maneuverability, and positioning accuracy.

A. Team Contributions

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- Resolved ROS2 environment and communication issues.
- Automation scripts (`ros_startup.sh` and `lazy_colcon.sh`) to simplify deployment.
- CLF-CBF-QP collision avoidance.
- Custom arm simulation and plots used in the report.
- Waypoint-based positioning strategy, to align near the goal.
- Maintained `All_Important_Commands.txt` reference document used throughout development.
- Report Section I (T0, T1, T5), Section II (T2, T3, T4)

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- Approximate linearization navigation controller.
- Integrated the arm and gripper control system.
- Added hockey Objects to the simulation.
- Stick pickup procedure using offsets.
- Implemented waypoint-based navigation.
- Rapid swing-shot maneuver used for goal scoring.
- Report Section I (T2, T3, and T4), Section II (T1, T5).

Joint Contributions: Experimental testing, system validation, debugging, final integration, and preparation of the Discussion section were completed collaboratively. Copilot was used for debugging assistance, LaTeX formatting, proof-reading, and technical writing support.

B. Additional info.

Videos, screenshots, source code, and project demonstrations are available in the project GitHub repository: [Click here for GitHub Repo.](#)